

Pandabox

Features and components

THE BOX



TOP VIEW



THE BOARD



BACK VIEW



FRONT VIEW



SIDE VIEW

PORTS

BACK SIDE

LCR PORT 2

LCR PORT 1

PRINTER PORT

LED
INDICATORS

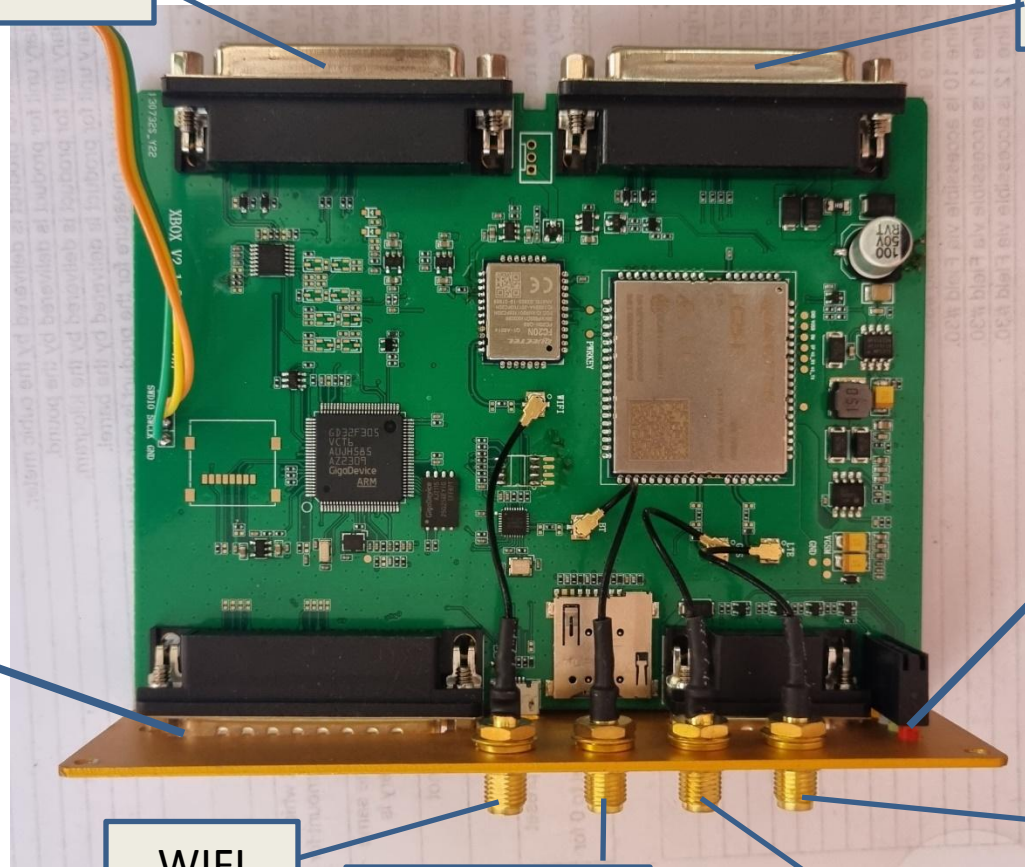
WIFI

BLUETOOTH

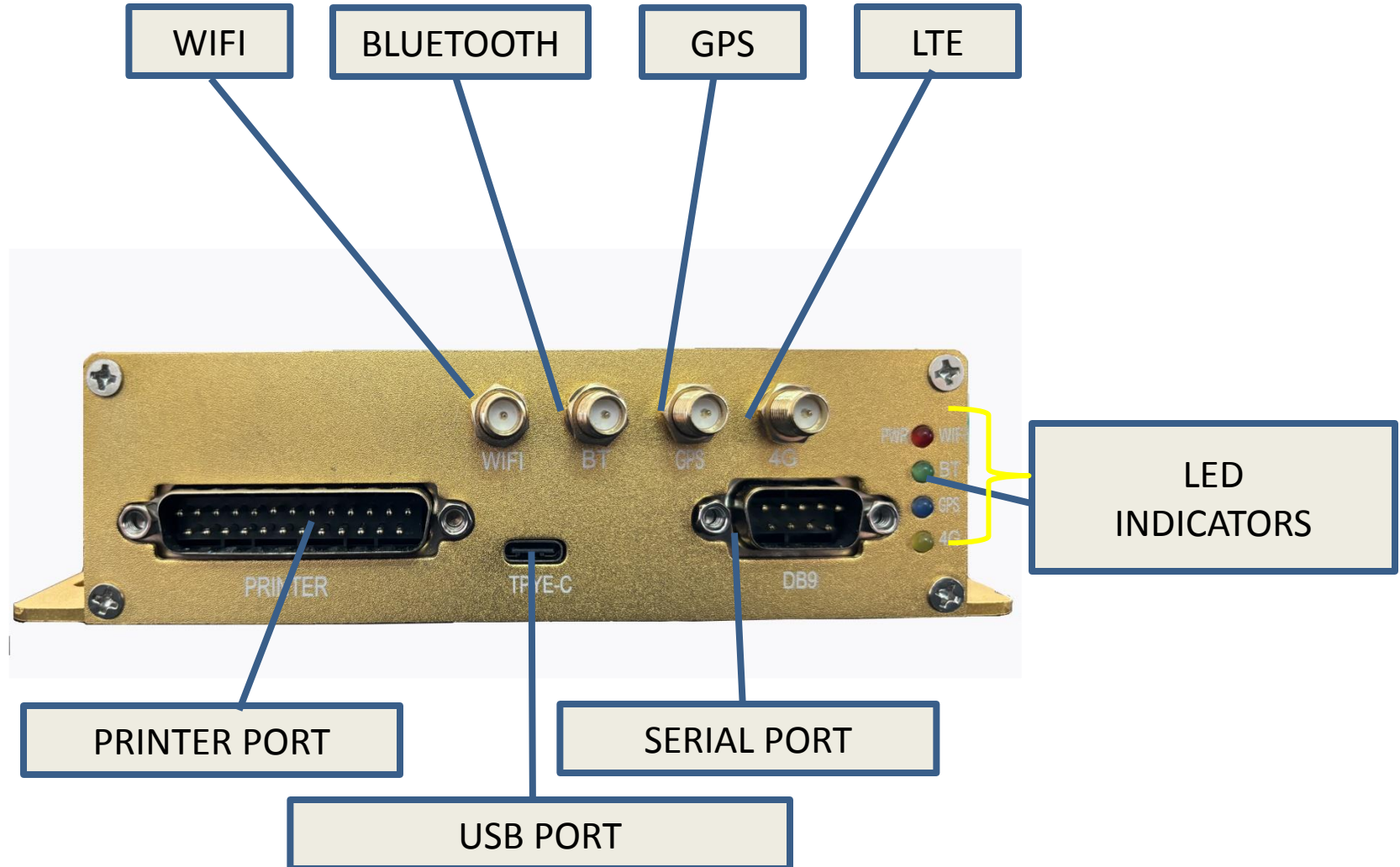
LTE

GPS

FRONT SIDE



FRONTSIDE PORTS



BACKSIDE PORTS



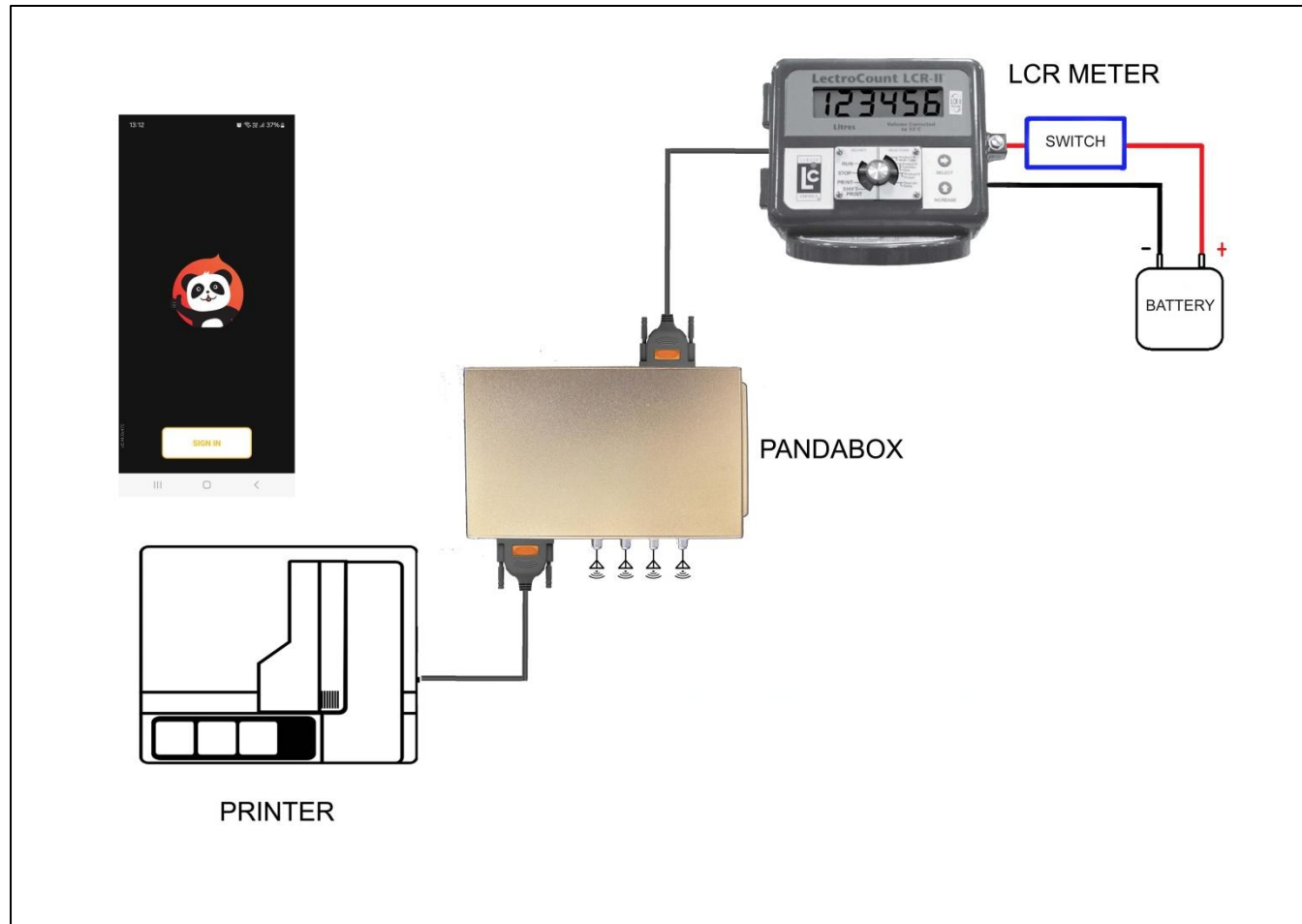
PORT 1
[CONNECT LCR
CABLE]

PORT 2
[CONNECT LCR
CABLE]

PORT 1 IS ON THE SAME SIDE AS THE
LED INDICATORS (FRONTSIDE)

PORT 2 IS ON THE SAME SIDE AS
THAT OF PRINTER PORT (BACKSIDE)

SYSTEM SETUP



ACCESSORIES

- Bluetooth and WiFi Antenna
 - A 10dBi antenna is provided with the box.
 - Customer can use any other higher powered antenna.
Check the connector end before buying one.



- GPS and LTE Antenna
 - Not function at the moment.



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POWERING THE BOX.

1. Power cable through dB9 port of the box and connected to truck's battery at other end hence provides 12V dc supply.
2. Powered through the LCR meter's power port. Derives 12 volt from the truck's battery again.



Since the power to the box is derived from the battery in the truck, power cycling of the box will require turning OFF the truck's ignition. Which may not be a wise thing to do during harsh weather conditions. Thus it is **recommended** to apply a switch to the LCR meters as well as the box.

COMMUNICATION MODES

Available modes

1. Bridge Mode (Mode 1):

It bypasses the box to communicate to the LCR meter. When assigned to bridge mode, the box **requires resetting of the LCR meter to function in mode 1**. So basically Mode 1 would enable manual mode of transaction. The mode 1 allows the user with following tasks.

- Pump and print (or simply, manual mode),
- Run Ezcommand tool,
- SDK demo (LC based apk for testing the LCR meter)

COMMUNICATION MODES

2. Data Mode/ Box Connection Mode (Mode 2)

When required to communicate via the box this mode is initiated. As the mobile App is connected to the box, the Mode 2 is activated. This allows the Pandabox to keep connected to the LCR node address of the LCR meter via LCP services.

Basically all the metered delivery is operated in Mode 2. With the Box connection mode ON, the metered delivery can be termed as;

LCR Metered Delivery

- Online: The assets are within the range of the Pandabox. The handheld device thus have continuous bluetooth connection with the box. The mode of delivery through the App can then be
 - NON-CONTINUOUS: Prints delivery ticket for individual asset on completion of the transaction. Thus resets the LCR meter after the completion of every delivery.
 - CONTINUOUS: The delivery ticket is printed after a set of assets are provided service to. Printing is not automatic, rather controlled by the mobile App or the User.

COMMUNICATION MODES

LCR Metered Delivery

- Offline: When the assets are outside the range of Pandabox such that the bluetooth connection gets lost, we call it the Offline mode. But with the Pandabox, we can still use the mobile App to capture the gallons delivered to any asset with the help of Pandabox. The box stores any change in gallons value from the LCR meter.
 - CONTINUOUS: During Offline mode, the LCR meter works in continuous mode such that the meter sends print command or resets only when the mobile App tells it to do so.

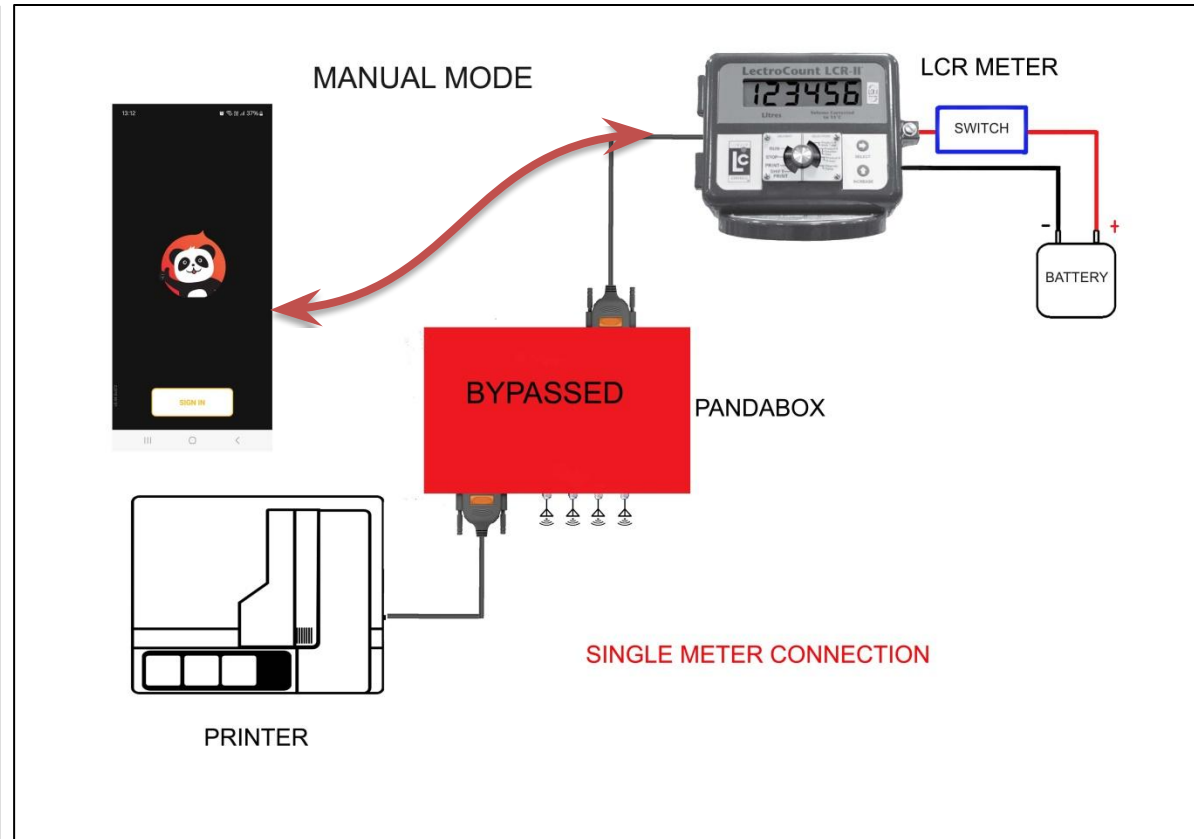
Note:

The gallons delivered during the Offline mode is stored in the box. Now to retrieve the data, the handheld device must again be in Online mode with the Pandabox. Once the bluetooth connection between the mobile App and Pandabox is retained, the gallons can be reflected to their respective assets.

COMMUNICATION MODES

MANUAL MODE (MODE 1)

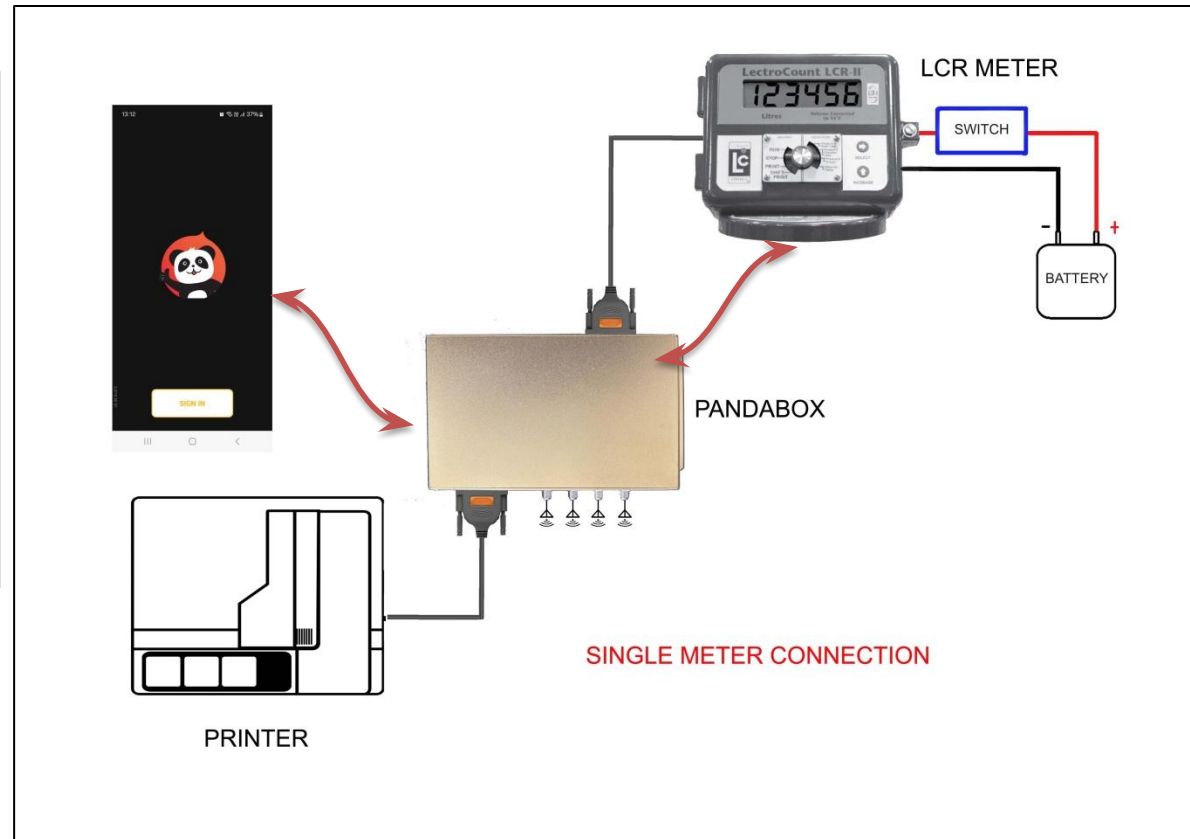
- During Manual mode, connection through Pandabox is bypassed such that the LCR meter can work independent of the Pandabox. The mobile App can also utilize the inbuilt bluetooth of LCRiQ meter to get connected to the meter as well.
- In order to terminate the manual mode we need to power cycle the LCR meter.
- We are allowed to connect to ezcommand tool in this mode only.
- We can also utilize liquid controls sdk demo App to connect to the LCR meter in this mode only.



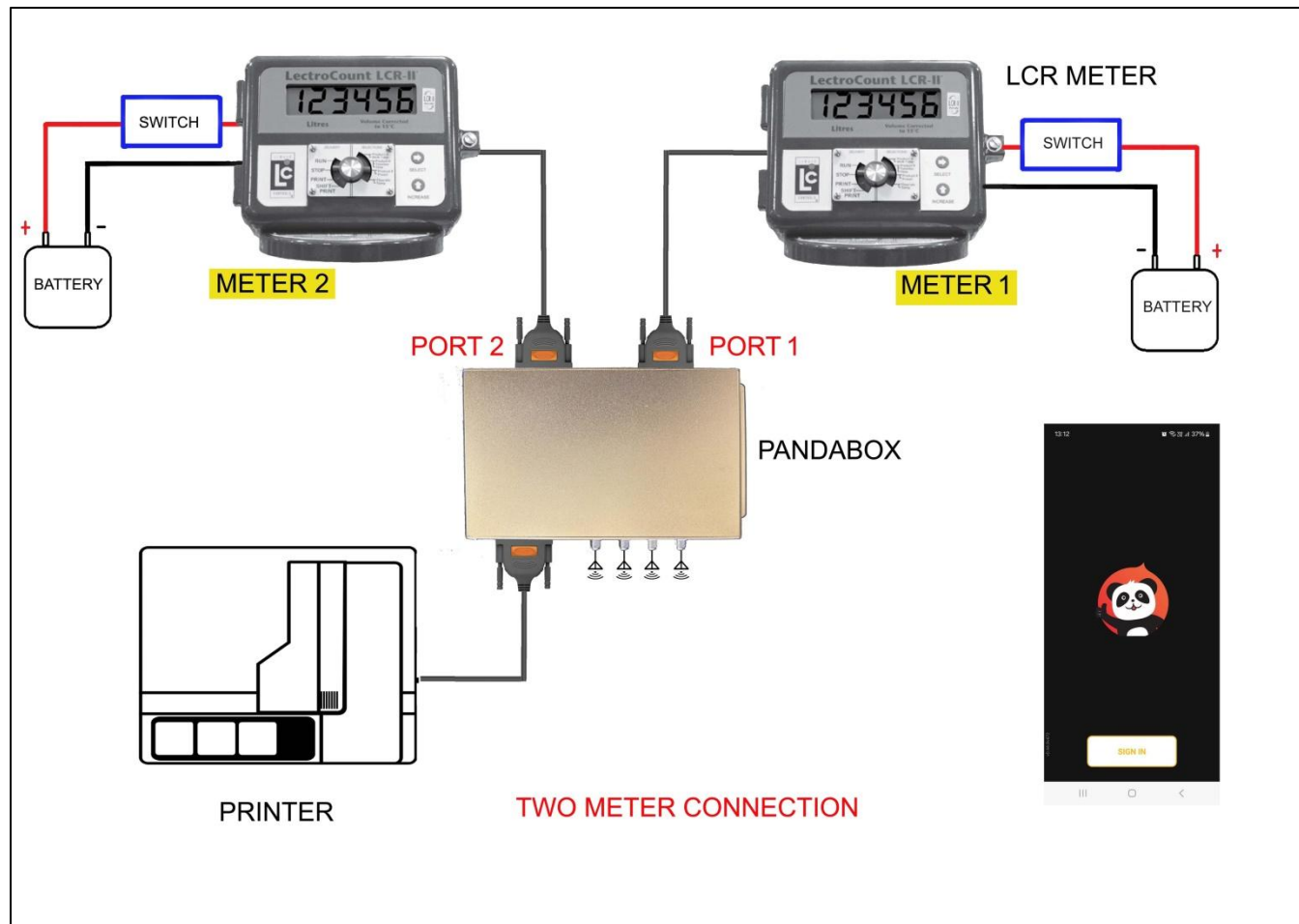
COMMUNICATION MODES

DATA MODE (MODE 2)

- DURING DATA MODE, THE MOBILE APP COMMUNICATES TO THE LCR METER VIA THE Pandabox.
- ANY INFORMATION TO BE SHARED FROM MOBILE APP TO THE LCR METER WILL GO THROUGH THE Pandabox, AND VICE VERSA.
- THE LCR METER WILL TOTALLY BE UNDER THE INFLUENCE OF THE APP.



TWO METER CONNECTIONS



PORTS AVAILABLE

1. BLUETOOTH - Functional with Fleetpanda Mobile App
 2. WIFI
 3. GSM
 4. GPS
- 
- currently not used. For future enhancements.
5. SERIAL (dB9 PORT) – Currently for PowerCURRENTLY FOR POWER
 6. PRINTER – supports printsharing functionality
 7. LCR PORTS FOR RS232 - for connecting LCR meters to the box.

PORTS AVAILABLE

BLUETOOTH –

- Connect the antenna provided with the box to the second SMA port from left. You can connect any other high power antenna but make sure of the connector end before purchasing one.
- By default the Box has bluetooth network name as 'Pandabrain' with the passkey '1234'.
- Connect and pair any handheld device with the box for communication.
- As for now, the fleetpanda App allows bluetooth communication.

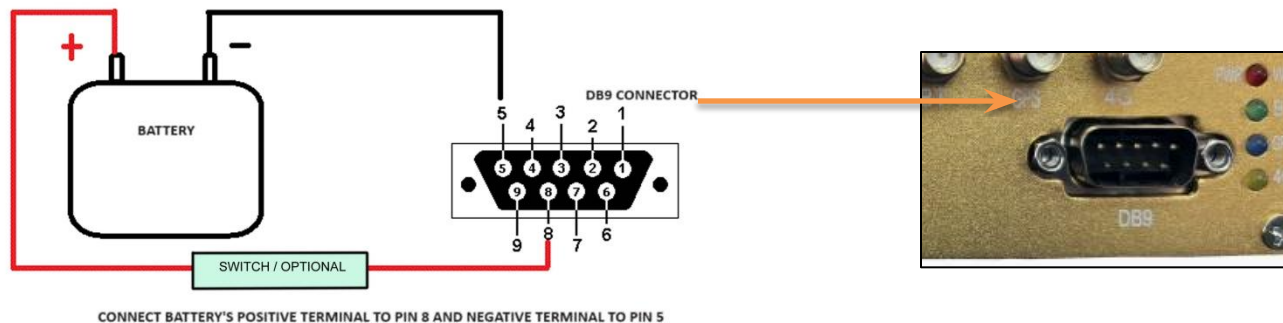


PORTS AVAILABLE

SERIAL (dB9 PORT)

- This port is currently dedicated to provide external power to the box. But it can also be used for serial communication for Ezcommand tool operation.
- It is also used for printing the mobile app information using Y-cables designed by Fleetpanda.

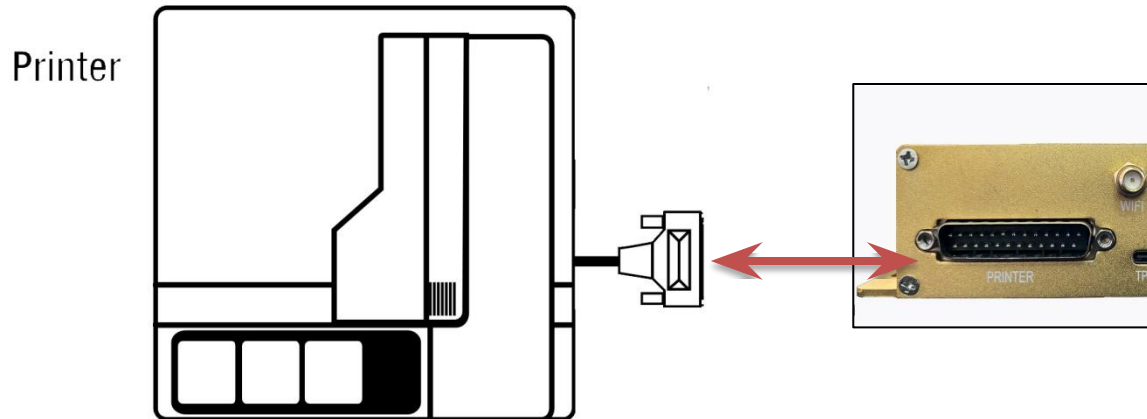
PROVIDING EXTERNAL POWER FROM DB9 PORT



PORTS AVAILABLE

PRINTER

- Connect the wired printer to the Printer Port available in the box. The box allows printshare capability such that either of the two LCR meters connected to the box can send the print command to the printer connected.



PORTS AVAILABLE

LCR PORTS

- There are currently two LCR ports available in the Box.
- We can thus connect two LCR meters to the box at any time.
- When connecting single meter to the box though we need to make sure that the meter is always connected to the LCR port 1.



CONNECTIVITY

1. RS232 SERIAL

- Printing, serial data communication

2. BLUETOOTH

- Current mode of communication between mobile app and pandabox.

3. WIFI

- Work in progress

4. GSM

5. GPS

NOT INTEGRATED TO APP AT THE MOMENT.